

POS Billing Application with Redux State Management

Overview

Build a React Native Point of Sale (POS) Billing Application that allows users to create invoices, manage store settings, generate receipt previews, and maintain billing history.

The objective of this assessment is to evaluate your understanding of React Native development, Redux state management, TypeScript, application architecture, and code quality.

Functional Requirements

1. Billing Module

Create a billing screen where users can:

- Add products to a bill
- Increase or decrease quantity
- Remove products from the bill
- Apply discounts
- Calculate GST/Tax
- Display subtotal and grand total

Sample Calculation

Coffee $\times$ 2 = ₹100

Tea $\times$ 1 = ₹20

Subtotal = ₹120

GST (5%) = ₹6

Total = ₹126

2. Receipt Preview

Generate a receipt preview after billing.

The receipt should contain:

- Store Name
- Invoice Number
- Date & Time
- Product List
- Quantity
- Tax Information
- Grand Total
- Footer Message

Sample Receipt

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STORE NAME

=====

Invoice : INV-1001

Coffee 2 x 50 100

Tea 1 x 20 20

Subtotal 120

GST 6

Total 126

Thank You

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Note:

Receipt formatting is mandatory.

Actual printer integration is optional.

3. Settings Module

Create a dedicated Settings screen.

Store Settings

- Store Name
- Store Address
- Phone Number

Billing Settings

- GST Enabled / Disabled
- GST Percentage
- Currency Symbol

Receipt Settings

- Footer Message
- Auto Print Toggle

All settings must be persisted locally and reflected throughout the application.

4. Billing History

Store completed invoices locally.

Display:

- Invoice Number
- Date
- Total Amount

Users should be able to:

- View invoice details
 - View receipt preview again
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Technical Requirements

Technology Stack

Mandatory:

- React Native
- TypeScript
- Redux Toolkit
- React Navigation

Local Storage:

- AsyncStorage or MMKV
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Redux State Management

Redux Toolkit must be used for managing application state.

Suggested Structure:

src/

|— store/

| |— index.ts

| |— billingSlice.ts

| |— settingsSlice.ts

| |— invoiceSlice.ts

Billing Slice

Manage:

- Cart Items
- Product Quantities
- Discounts
- Tax Calculations
- Bill Totals

Settings Slice

Manage:

- Store Information
- Billing Configuration
- Receipt Configuration

Invoice Slice

Manage:

- Invoice History
 - Current Invoice
 - Invoice Details
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Expected Screens

1. Billing Screen

- Product List
- Cart Summary
- Totals Section
- Generate Bill Action

2. Receipt Preview Screen

- Printable Receipt Layout

3. Billing History Screen

- Invoice List
- Invoice Details

4. Settings Screen

- Store Settings
 - Billing Settings
 - Receipt Settings
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Project Structure

Applicants are expected to follow a scalable folder structure.

Example:

src/

├─ screens/

├─ components/

├─ navigation/

├─ store/

├─ services/

├─ hooks/

├─ types/

├─ utils/

└─ assets/

Deliverables

Candidates must submit:

1. GitHub Repository
 2. README Documentation
 3. APK Build
 4. Application Screenshots
 5. Architecture Overview
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